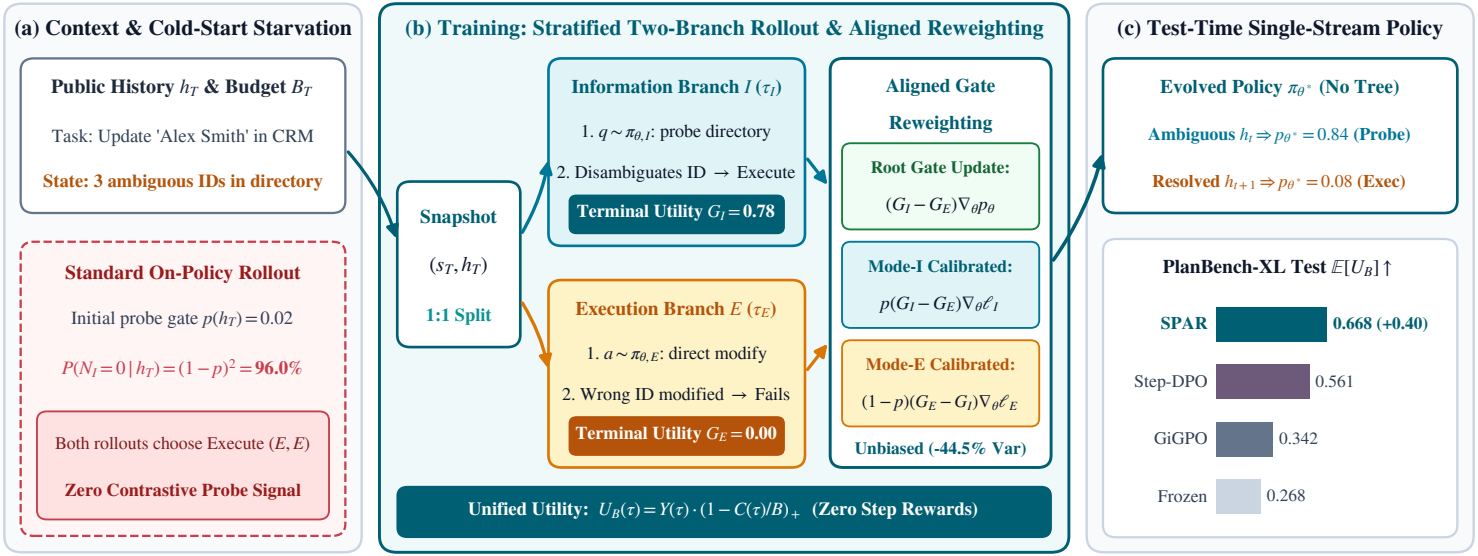
(a) Context & Cold-Start Starvation
Public History h_T & Budget B_T
Task: Update 'Alex Smith' in CRM
State: 3 ambiguous IDs in directory
Standard On-Policy Rollout
Initial probe gate $p(h_T) = 0.02$
$P(N_I = 0 \mid h_T) = (1-p)^2 = \mathbf{96.0\%}$
Both rollouts choose Execute (E, E)
Zero Contrastive Probe Signal
(b) Training: Stratified Two-Branch Rollout & Aligned Reweighting
Snapshot
(s_T, h_T)
1:1 Split
Information Branch I (τ_I)
1. $q \sim \pi_{\theta,I}$: probe directory
2. Disambiguates ID → Execute
Terminal Utility $G_I = 0.78$
Execution Branch E (τ_E)
1. $a \sim \pi_{\theta,E}$: direct modify
2. Wrong ID modified → Fails
Terminal Utility $G_E = 0.00$
Aligned Gate Reweighting
Root Gate Update:
$(G_I - G_E)\nabla_\theta p_\theta$
Mode-I Calibrated:
$p(G_I - G_E)\nabla_\theta \ell_I$
Mode-E Calibrated:
$(1-p)(G_E - G_I)\nabla_\theta \ell_E$
Unbiased (-44.5% Var)
Unified Utility: $U_B(\tau) = Y(\tau) \cdot (1 - C(\tau)/B)_+$ (Zero Step Rewards)
(c) Test-Time Single-Stream Policy
Evolved Policy π_{θ^*} (No Tree)
Ambiguous $h_t \Rightarrow p_{\theta^*} = 0.84$ (Probe)
Resolved $h_{t+1} \Rightarrow p_{\theta^*} = 0.08$ (Exec)
PlanBench-XL Test $\mathbb{E}[U_B] \uparrow$
SPAR
0.668 (+0.40)
Step-DPO
0.561
GiGPO
0.342
Frozen
0.268